



Project Details

1. Project Title

EntraLens – Intelligent Identity Security Copilot for Microsoft Entra ID

AI-powered Identity Security Dashboard with Natural Language Querying, Voice Assistant, SQL Generation, User Management, Risk Analytics, and Identity Governance.

2. Problem Statement

Large organizations use Microsoft Entra ID to manage thousands of users, licenses, privileged accounts, and security events.

Challenges:

- Inactive accounts remain enabled
- MFA compliance is difficult to monitor
- Privileged accounts create security risks
- License wastage increases costs
- Administrators need SQL knowledge to analyze data
- Security investigations are time-consuming

The goal is to provide an AI-powered security copilot that helps administrators monitor, investigate, and manage identity security using natural language and voice commands.

3. Target Users

Primary Users

- IT Administrators
- IAM Engineers
- Azure Cloud Engineers
- Security Analysts

- SOC Teams
- Compliance Teams

Secondary Users

- Enterprise Security Managers
- Global Administrators
- Billing Administrators

4. Core Features

Identity Monitoring

- Inactive Users
- MFA Compliance
- Privileged Accounts
- License Optimization
- Orphan Accounts
- Conditional Access
- Identity Governance
- Access Review

User Management

- Create User
- Read User
- Update User
- Delete User

Dashboard Analytics

- KPI Cards
- Risk Score
- AI Summary
- Charts
- Investigation Output

Reporting

- CSV Export
- PDF Export

5. AI Features

AI Security Copilot

User asks:

Show inactive users

AI analyzes and responds.

Voice Assistant

User speaks:

Show high risk users

AI:

- Converts speech to text
- Detects intent
- Generates SQL
- Runs query
- Returns result

Intent Detection

Example:

Finance department-এর user দেখাও

Detected Intent:

Department Search

AI SQL Generator

Generated Query:

```
SELECT *  
FROM users  
WHERE LOWER (department)=LOWER('Finance');
```

Bilingual AI

Supports:

- English
- Bangla

6. RAG Architecture

Current RAG Flow:

User Question



AI Intent Detection



Knowledge Context



SQL Generation



Database Retrieval



Response Generation

Future Enhancement:

PDF

Policies

Audit Reports

Security Documents



Vector Database



Semantic Search



AI Response

7. Live Database Query Architecture

User



AI Copilot



Intent Detection



SQL Generation



Fast API



PostgreSQL



Query Execution



Results

Example:

Show Finance Users



```
SELECT *  
FROM users  
WHERE LOWER (department)=LOWER('Finance');
```



Results Returned

8. Vision AI Use Case

Vision AI Module (Future Enhancement)

Identity Verification

Upload user image



Face Detection



Identity Matching



Security Verification

Possible Use Cases

- Face Recognition Attendance
- Identity Verification
- Access Control
- Visitor Monitoring

9. Local / Cloud Model Decision

Local AI

Platform:

Ollama

Benefits:

- No data leaves organization
- Better privacy
- Lower cost

Models:

- Gemma
- Llama
- Qwen

Cloud AI

Benefits:

- Better reasoning
- Higher accuracy
- Enterprise scalability

Current Project:

Hybrid Architecture

Local + Cloud Ready

10. Database Schema

The EntraLens platform uses PostgreSQL as the primary database for storing identity information, role assignments, security logs, and audit events. The following tables are used within the system.

Table 1: Users

Column Name	Data Type	Key	Description
id	INTEGER	PK	Unique user identifier
tenant_id	VARCHAR(50)	-	Tenant identifier
display_name	VARCHAR(100)	-	User full name
email	VARCHAR(150)	UNIQUE	User email address
department	VARCHAR(100)	-	User department
user_type	VARCHAR(50)	-	User category
mfa_enabled	BOOLEAN	-	MFA status (True/False)
license_type	VARCHAR(20)	-	Assigned license type
last_login_days	INTEGER	-	Number of days since last login
risk_score	INTEGER	-	Calculated security risk score
account_status	VARCHAR(20)	-	Active / Terminated

Table 2: Role Assignments

Column Name	Data Type	Key	Description
id	INTEGER	PK	Unique role assignment ID
user_email	VARCHAR(150)	FK	Associated user email
role_name	VARCHAR(100)	-	Assigned role
privileged	BOOLEAN	-	Indicates privileged access
assigned_date	DATE	-	Date of role assignment

Table 3: Sign-In Logs

Column Name	Data Type	Key	Description
id	INTEGER	PK	Unique log identifier
user_email	VARCHAR(150)	FK	User email
login_time	TIMESTAMP	-	Login timestamp
ip_address	VARCHAR(50)	-	Source IP address
country	VARCHAR(100)	-	Login location
status	VARCHAR(20)	-	Login status (Success/Failed)

Table 4: Audit Events

Column Name	Data Type	Key	Description
id	INTEGER	PK	Unique event identifier
event_type	VARCHAR(100)	-	Event category
user_email	VARCHAR(150)	FK	Associated user
description	TEXT	-	Event details
created_at	TIMESTAMP	-	Event creation timestamp

Database Relationships

Parent Table	Child Table	Relationship
Users	Role Assignments	One-to-Many
Users	Sign-In Logs	One-to-Many
Users	Audit Events	One-to-Many

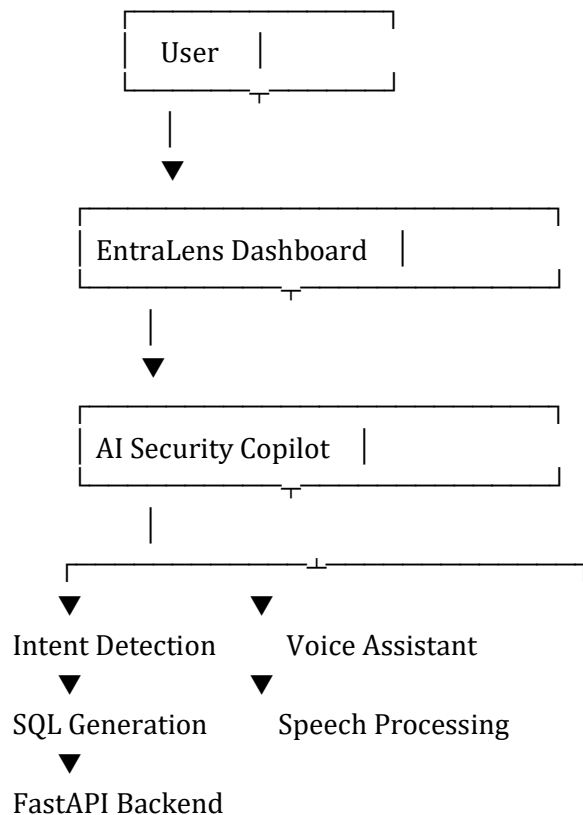
Database Purpose

The database serves as the core data source for:

- AI Intent Detection
- Natural Language to SQL Query Generation
- Identity Security Analytics
- User Management (CRUD)
- MFA Compliance Monitoring
- Inactive User Detection
- Risk Assessment
- Role-Based Access Control (RBAC)
- Security Investigation Workflows

The AI engine retrieves and analyzes information from these tables to generate real-time security insights and SQL query results for administrators.

11. System Architecture Diagram



▼
PostgreSQL Database
▼
Dashboard Results

12. Demo Workflow

Scenario 1

User asks:

Show inactive users

↓

AI detects intent

↓

SQL generated

↓

Results displayed

Scenario 2

User speaks:

High risk user কারা?

↓

Voice Recognition

↓

Intent Detection

↓

SQL Generation

↓

Result Display

Scenario 3

Admin adds a new user

↓

Database Updated

↓

Dashboard KPI Updated

↓

Risk Recalculated

13. Security and Permission Plan

Global Administrator

Full Access

IAM Engineer

- User Management
- Identity Governance

SOC Analyst

- Dashboard
- Security Monitoring

Billing Administrator

Only:

- License Optimization
- Orphan Accounts
- Inactive Users

Unauthorized Users

Access Denied

14. Tech Stack

Frontend

- HTML
- CSS
- JavaScript
- Chart.js

Backend

- Python 3.13
- FastAPI
- SQLAlchemy
- Uvicorn

Database

- PostgreSQL

AI

- Ollama
- Gemma / Llama / Qwen

Voice

- Web Speech API

Version Control

- Git
- GitHub

15. Future Improvement Plan

Phase 2

- Real Microsoft Graph API Integration
- Azure Entra ID Integration
- Real-Time Security Monitoring
- Multi-Tenant Support

AI Improvements

- Advanced RAG
- Vector Database
- Security Knowledge Base
- Threat Intelligence Integration

Vision AI

- Face Recognition
- Identity Verification
- Visitor Access Management

Enterprise Features

- Multi-Agent AI
- SIEM Integration
- Sentinel Integration
- Automated Incident Response